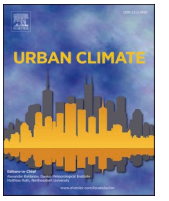
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Hyperlocal air pollution in an urban environment - measured with low-cost sensors

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ABSTRACT

Air pollution levels can vary significantly over short distances, particularly in urban areas and near emission sources. This study examined the performance of low-cost sensor devices for monitoring levels of NO_2, O_3, and $PM_{2.5}$ along two closely spaced (average 8 m) routes in Copenhagen, Denmark. One route was located near a lake (Route 1) and the other near a busy road (Route 2). The routes were walked in tandem for 84 h. The mode of deployment was determined using an accelerometer, gyroscope, and light sensor, achieving a 97.4 % accuracy rate. Field calibration with multivariate linear regression proved the most robust calibration model across pollutants, yielding mean R^2-values of 0.64, 0.79, and 0.48 for NO_2, O_3, and $PM_{2.5}$, respectively. The sensor intervariability was generally low, with mean R^2-values of 0.84–0.94 for $PM_{2.5}$ measured with optical particle sensors and 0.88–0.90 for NO_2 and O_3 measured with metal-oxide sensors. Results showed significantly higher NO_2 concentrations on Route 2 (21.6 ± 6.6 ppb) compared to Route 1 (10.1 ± 4.0 ppb) during mornings. However, no significant differences in O_3 and $PM_{2.5}$ concentrations were observed. Our findings demonstrate that low-cost sensors can accurately quantify air pollution exposure in urban areas with high spatiotemporal resolution.

1. Introduction

Air pollution is a well-documented health hazard, with particulate matter (PM) with an aerodynamic diameter smaller than 2.5 μm ($PM_{2.5}$), nitrogen dioxide (NO_2), and ground-level ozone (O_3) classified as the most harmful urban air pollutants for human health (Schraufnagel et al., 2019; Sicard et al., 2023). Exposure to these pollutants has been linked to various acute and chronic health issues, such as the increased risk of cardiovascular disease and respiratory infections, dementia, Parkinson's disease, cognitive impairment, kidney disease, diabetes, and issues with the reproductive system (Guarnieri and Balmes, 2014; Franklin et al., 2015; Al-Aly and Bowe, 2020; Yang et al., 2020; Curtis et al., 2006; Zhang et al., 2018; Pourvakhshoori et al., 2020).

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The World Health Organization recently tightened its air quality exposure guidelines, setting the annual mean exposure guideline for NO_2 at 10 μg m^{-3} (~5 ppb), the 8-h mean for peak season exposure guideline for O_3 at 60 μg m^{-3} (~30 ppb), and the annual mean exposure guideline for $PM_{2.5}$ at 10 μg m^{-3} (WHO, 2021). The European Union has also announced its intention to revise its ambient air quality directive (2008/50/EC) (European Commission, 2020).

Air quality monitoring has traditionally relied on high-precision instruments at static air quality monitoring stations (AQMSs) (Motlagh et al., 2020). However, the instruments in these AQMSs are generally expensive to buy and costly to maintain. A network of the stations also requires additional infrastructure including laboratories for calibrating instruments and weighing gravimetric samples, as well as the employment of specialized personnel. Because of these factors, there is a limited number of fixed AQMSs, leading to sparse coverage (Motlagh et al., 2020; Kumar et al., 2015; Maag et al., 2018).

Field studies have shown that pollution levels can vary rapidly within urban environments, depending on location, time of day, week, and season (Frederickson et al., 2022; Borge et al., 2019; Apte et al., 2017; Marshall et al., 2008; Karner et al., 2010; Steinle et al., 2013). Moreover, individual exposure can vary depending on specific time-activity patterns and micro-environments (Vrijheid, 2014; Wild, 2005; Mayer, 1999). However, exposure estimates based on current fixed AQMSs are not able to take these factors into account (Yu et al., 2018). They do not represent the air quality in indoor environments (Van Roosbroeck et al., 2008; Galatioto et al., 2021), at hot spots such as within street canyons (Vardoulakis et al., 2003), or while commuting (Frederickson et al., 2020a). Indoor exposure is especially important since it is estimated people spend ~90 % of their time inside (Cincinelli and Martellini, 2017; Wild, 2012).

To address these limitations, low-cost sensor (LCS) devices are increasingly being used to increase the spatial resolution of air pollution monitoring. Although these sensors are less accurate than reference-grade instruments, they are cost and size efficient, making them suitable for use in dense monitoring networks (Barcelo-Ordinas et al., 2019; Mead et al., 2013; Morawska et al., 2018; Popoola et al., 2018), mobile monitoring (Chatzidiakou et al., 2019; Genikomsakis et al., 2018; Lim et al., 2019; Gao et al., 2016; von Schneidemesser et al., 2019; Lin et al., 2017), and personal exposure assessment (Piedrahita et al., 2014; Mahajan and Kumar, 2020). Wearable sensors combined with GPS tracking can estimate spatial distributions of air pollution in the different micro-environments and provide personalized estimates of air pollution exposure, complementing the sparsely located AQMSs and/or modeled address-level data (Russell et al., 2022a; Sloan et al., 2016). Although personal gravimetric samplers can be used for PM measurements, and diffusion tubes for gas measurements, with greater accuracy than low-cost sensors, they can only measure at low time resolution.

While low-cost sensors have known problems with data quality due to issues with selectivity, stability, and sensitivity (Karagulian et al., 2019), their use can still provide valuable information (Frederickson et al., 2020b). For some monitoring objectives, reaching the same level of accuracy as reference instruments may not always be necessary. For instance, in highly variable environments the benefits of dense networks or mobile monitoring may outweigh the lower data quality of low-cost sensors compared with high-quality AQMSs at limited points. Low-cost sensors can provide insights into otherwise unknown details in the human environment and can help determine exposure based on the path integral (Nagendra et al., 2019). The additional information obtained can provide supplementary data to the reference monitoring by the AQMSs (Snyder et al., 2013). However, it is crucial to know the exact level of accuracy and performance characteristics of the low-cost sensors when utilizing the data (Snyder et al., 2013). Utilizing a large number of low-cost sensors simultaneously and comparing their data relative to each other can also significantly enhance confidence in measurements, as it increases the sample size.

Each type of sensor has its own issues that affect the measurement accuracy. In addition, there is often variability between different low-cost sensors of the same type due to their low-cost production methods and each sensor's individual history (Austin et al., 2015; Sousan et al., 2017). Developing calibration methods and models as well as corrections to convert the raw sensor response to concentration is integral to their use. It is crucial that device-specific calibration is conducted against reference-grade instruments in an environment that is representative of the place the low-cost sensor will be used (Maag et al., 2018). This is because calibration factors may vary for different devices of the same type and model, and for the same device under different conditions (Morawska et al., 2018). Especially temperature and relative humidity have been shown to affect the performance of the low-cost sensors (Masson et al., 2015; Crilley et al., 2018; Crilley et al., 2020). Therefore, frequent testing and (re-)calibration of the low-cost sensors is important and may help overcome some of these issues (Spinelle et al., 2017; Rai et al., 2017).

The aim of this study is to evaluate the effectiveness of LCS nodes for measuring air pollution levels within a specific micro-environment in Copenhagen, Denmark, with a fine spatiotemporal resolution. Specifically, we measured the concentrations of NO_2, O_3, and $PM_{2.5}$, which are known to be harmful to human health. The study aims to assess whether LCS nodes can detect differences in air pollution levels between two routes that are separated by an average of approximately 8 m. The results of this study can provide valuable insights into the use of LCS nodes for personal exposure monitoring and can help to assess the feasibility of using such devices in future studies. Additionally, the results can provide information on the spatial and temporal variations in air pollution levels in the specific micro-environment studied and can help to identify potential hot spots of air pollution in the area.

2. Materials and methods

2.1. Campaign design and study area

Copenhagen, a Scandinavian city known for its high rates of active mobility like walking, jogging, and cycling, is also confronted with a significant number of private cars. As of 2021, there were 134,880 private cars in Copenhagen, with 51,400 (38.1 %) being diesel vehicles (Danmarks Statistik, 2022), a known source of NO_2 and $PM_{2.5}$. A distinctive feature of Copenhagen's topography is the row of three rectangular lakes curving around the western margin of the Copenhagen city center, see Fig. 1.

In this work, a field campaign was conducted around the northern part of Black Pond Lake, as shown on the map over Copenhagen

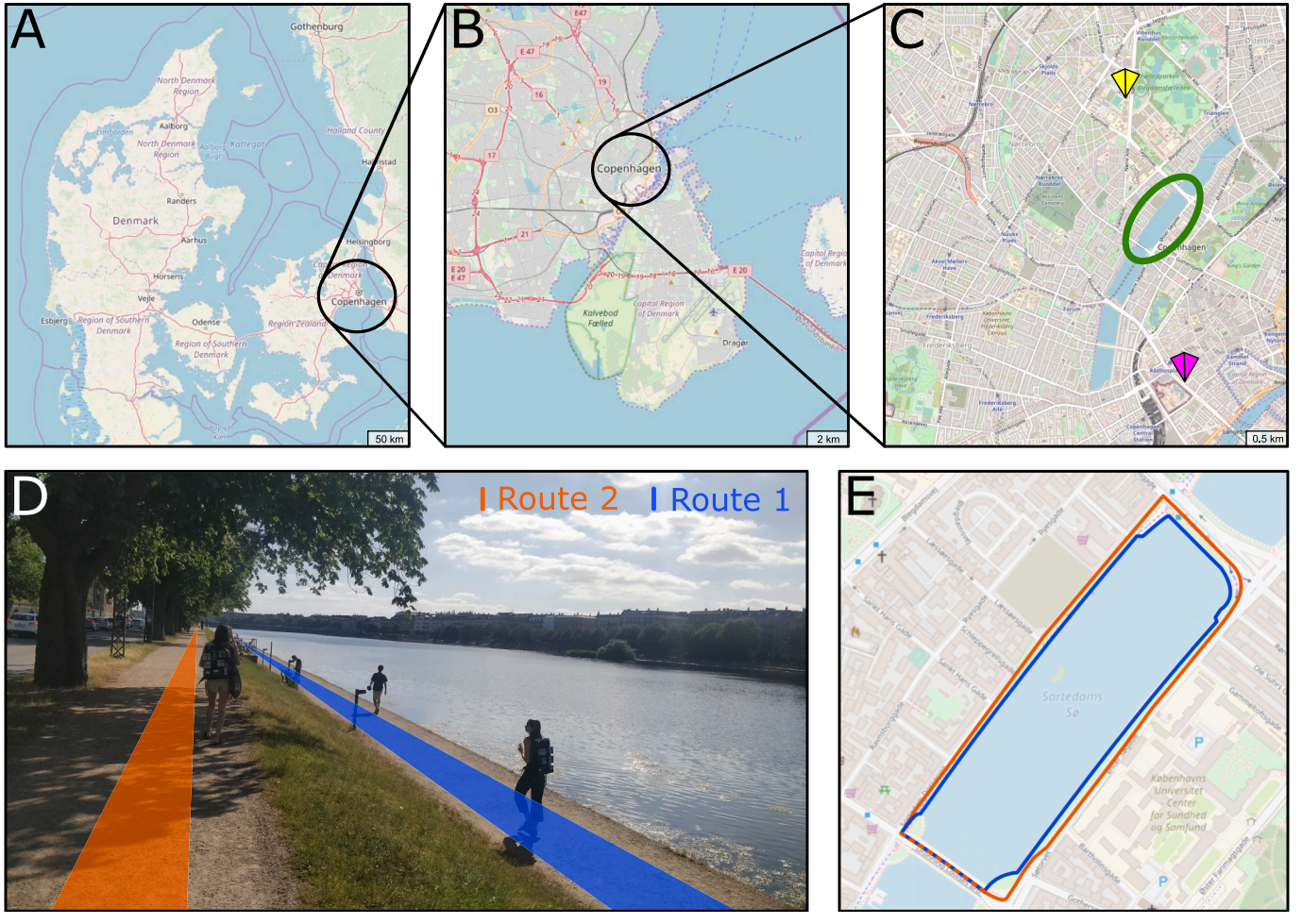


Fig. 1. Map over Denmark (A), Copenhagen (B), and the campaign area (C). Within the campaign area, the Black Pond Lake is marked with a green oval, the rooftop urban background station (HCØ) with a yellow marker, and the urban roadside station (HCAB) with a pink marker. Street view (D) and top view (E) over the two variations of the route, where Route 1 (blue) is closest to the lake, and Route 2 (orange) is closest to the road. The maps are obtained from © OpenStreetMap contributors (OpenStreetMap contributors, 2021). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in Fig. 1C. The lake is bordered on three sides by highly-trafficked roads, namely Øster Søgade to the East and two bridges - Queen Louise's Bridge to the South and the Bridge of Peace to the North. The fourth side borders a residential area with several cafes along Sortedam Dossering to the West. Two people walked the two variations of the 1.7 km route around the lake in tandem. Route 1 was situated closest to the lake, while the other route, Route 2, was closer to the road. The two routes were separated with an average distance of approximately 8 m and a range of 1 to 30 m. It should also be noted that Route 2 is at the same elevation as the road, while Route 1 is approximately 1 m lower. Each person carried a backpack equipped with LCS nodes that measured levels of NO_2, O_3, and $PM_{2.5}$ as well as acceleration, angular velocity, and other environmental factors such as temperature, relative humidity, and light levels.

The campaign was carried out for five consecutive weeks between May 24 and June 23, 2021. Two time windows during weekdays were chosen for monitoring, based on known rush-hour traffic times. Measurements began at 07:00 ('morning rush hour') and 17:00 ('evening rush hour') and lasted two hours each. Data collection took place in Copenhagen in May and June 2021, when society was mostly re-opened after the lockdown in the spring of 2021 due to the COVID-19 pandemic. Total traffic counts in Denmark were comparable to those in 2019 (before the COVID-19 pandemic), with a small reduction in traffic index in May (4.2 %) and a small increase in traffic index in June (3.1 %), compared to the same months in 2019 (Vejdirektoratet, 2022).

2.2. Low-cost sensor nodes

Each volunteer carried a backpack equipped with several LCS nodes and a smartphone for GPS tracking and WiFi connection for the LCS nodes. Two types of LCS nodes were used in this study: Gen4P nodes from DevLabs and PEM nodes from ScioSense.

The Gen4P nodes are weatherproof with dimensions of 88 × 88 × 90 mm, and include a metal oxide (MOx) sensor for measuring NO_2 and O_3 and an optical PM sensor for $PM_{2.5}$. The housing contains inlets for the low-cost sensors and a fan that ensures the air is actively and steadily sampled. Each Gen4P node is connected to WiFi allowing transmission of the raw data to a cloud-based database with a 1-min time resolution.

The PEM nodes are 115 × 70 × 30 mm and record to a microSD card via an Adalogger FeatherWing with a 1-min time resolution. They consist of an optical PM sensor and a number of additional sensors for the measurement of movement and environmental conditions. These nodes are also equipped with a fan, allowing active sampling.

All nodes are powered with USB cables connected to external power banks. Table 1 lists the specific types of low-cost sensors in the Gen4P and PEM nodes. The MOx and optical PM sensors are described below.

2.2.1. MOx sensor: MiCS-6814

The MOx sensor used is the MiCS-6814 from SGX Sensortech, which is also described in detail in Russell et al. (2022a). The detection limit of this sensor is ~1 ppb, a full list of specifications can be found in this data sheet (SGX Sensortech, 2015). Each sensor consists of (among others) two independent sensor chips with sensing layers and resistors. The sensor chips are denoted s_2 for reducing gases and s_3 for oxidizing gases. The principle of operation is based on measuring the change in conductance when the device is exposed to its target gas, which interacts with the sensor chips' surface. Within the Gen4P node, there are two MOx sensors separated by an O_3 filter. Thus, the first sensor ($M1$) is exposed to both gases whereas the second sensor ($M2$) is only exposed to NO_2 (Viricelle et al., 2006). The sensor chips are operated with temperature cycling, in which two voltage levels are applied alternately to a resistive heater embedded in the sensor chips for 30 s each. The gas of interest will preferentially bind to the surface at a specific point in the temperature cycle. During the high-temperature period, contaminants will be burnt off leaving a clean surface. For each sensor chip, scale and offset values are recorded for both the high- and low-temperature segments of the cycle, and the mean conductance during the two temperature periods is also recorded.

2.2.2. Optical PM sensors: SPS30 and SDS011

The PM sensors, SPS30 from Sensirion and SDS011 sensor from Nova Fitness operate using the light scattering principle, and their sensor specifications can be found in Sensirion (2020) and Nova Fitness (2015). An algorithm is used to calculate the mass concentration for PM by using the particle number concentration measured, the particle size distribution, and an assumed particle density (Liu et al., 2019). The sensor module includes a fan to ensure a continuous flow of air through the sensor chamber (Genikomsakis et al., 2018). The sensors can measure the particle density distribution between 0.3 and 10 μm, and the $PM_{2.5}$ detection range for both sensors is 0 to 999 μg m^{-3} (Budde et al., 2018). The main difference between the SDS011 and SPS30 sensors is the HEPA filter within SPS30,

Table 1
Low-cost sensors used within this study, together with their output(s) after calibration. Abbreviations: MOx = metal oxide, RH = relative humidity, T = temperature, and ch$X$ = channel number X.

Sensor	Manufacturer	Type	Output	~Price	Node
MiCS-6814	SGX Sensortech	MOx	NO_2, O_3 / ppb	€12	Gen4P
SPS30	Sensirion	Optical	$PM_{2.5}$ / μg m^{-3}	€44	PEM
SDS011	Nova Fitness	Optical	$PM_{2.5}$ / μg m^{-3}	€35	Gen4P
ENS210	ScioSense	Relative humidity, temperature	T/ °C, RH/ %	€2	PEM
TCS3430	ScioSense	Colorimeter	n_{ch0}, n_{ch1}, n_{ch2}, n_{ch3}	€4	PEM
BMI270	Bosch Sensortech	Accelerometer, gyroscope	Acceleration, orientation	€5	PEM

which is used to create sheath air.

2.3. Air quality monitoring stations

The data obtained from the measurement campaign is compared with data from two official AQMSs operated under the air pollution part of the NOVANA (Danish National Monitoring Program for Water and Nature) program (Ellermann et al., 2020). The first AQMS is the rooftop urban background station at H.C. Ørsted Institute (HCØ; 55.700139°N, 12.561318°W; 1.3 km from the measurement area) at the University of Copenhagen and the second is the urban roadside station at H. C. Andersen's Boulevard (HCAB; 55.674318°N, 12.571100°W; 1.7 km from the measurement area) in central Copenhagen. The location of these AQMSs is shown in Fig. 1C. Both HCAB and HCØ have a low volume sampler (LVS) for daily average measurements of $PM_{2.5}$ as well as reference-grade instruments using chemiluminescence and UV absorption for measurements of NO_2 and O_3 at 30-min time resolution, respectively. In addition, HCAB has 30-min measurements of $PM_{2.5}$ using tapered element oscillating microbalance (TEOM) analyses. Moreover, HCØ has a meteorological station that measures temperature, relative humidity, irradiance, wind speed, and wind direction at a 30-min time resolution.

2.4. Calibration and data analysis

The analyses were conducted in R version 3.6.1 (via *Rstudio*). The LCS nodes were calibrated by co-locating them at HCAB for one month (June 24 to July 24, 2021) and comparing the LCS data with the reference data obtained from the AQMS. The calibration models for each LCS were trained on the first three weeks of data, and the last week was used to validate the calibration models. Calibration models for the LCS measurements were built using linear regression (LR), multivariate linear regression (MLR), and artificial neural network (ANN). To build the ANN model, detrended, normalized data was used as input in the neuralnet function in R. Three hidden neurons were used per layer, and the other parameters were set to default. The input variables for the different calibration models for each sensor type are as described below. Temperature (T) and relative humidity (RH) were only included in the MLR and ANN models. For each MOx sensor, the input variables for the prediction of NO_2 and O_3 were the following:

$$\begin{aligned} P_{\mathrm{NO_2\ or\ O_3}} = a_0 + \\ a_1 \cdot M1_s_2_\mathrm{high} + a_2 \cdot M1_s_2_\mathrm{low} + a_3 \cdot M1_s_2_\mathrm{offset} + a_4 \cdot M1_s_2_\mathrm{scale} + \\ a_5 \cdot M1_s_3_\mathrm{high} + a_6 \cdot M1_s_3_\mathrm{low} + a_7 \cdot M1_s_3_\mathrm{offset} + a_8 \cdot M1_s_3_\mathrm{scale} + \\ a_9 \cdot M2_s_2_\mathrm{high} + a_{10} \cdot M2_s_2_\mathrm{low} + a_{11} \cdot M2_s_2_\mathrm{offset} + a_{12} \cdot M2_s_2_\mathrm{scale} + \\ a_{13} \cdot M2_s_3_\mathrm{high} + a_{14} \cdot M2_s_3_\mathrm{low} + a_{15} \cdot M2_s_3_\mathrm{offset} + a_{16} \cdot M2_s_3_\mathrm{scale} \\ (+ a_{17} \cdot T + a_{18} \cdot RH) \end{aligned} \tag{1}$$

where a_0 is the offset, a_1-a_{16} are the calibration coefficients, and a_{17} and a_{18} are the temperature and relative humidity correction coefficients, calculated separately for each MOx sensor. $M1$ and $M2$ are the two MOx sensors within the Gen4P node, which both include the two sensor chips (s_2 and s_3). s_2 was included in the model to check for interference.

For prediction of $PM_{2.5}$, the sensor response ($PM_{2.5}$) is used directly as input:

$$P_{\mathrm{PM_{2.5}}} = a_0 + a_1 \cdot PM_{2.5} (+ a_2 \cdot T + a_3 \cdot RH) \tag{2}$$

where the offset (a_0), sensitivity (a_1), and correction coefficients (a_2 and a_3) are determined separately for each PM sensor.

The data validation was quantified with the statistical indicators: Correlation of determination (R^2), slope, intercept, root-mean-square error (RMSE), mean bias error (MBE), and mean absolute error (MAE). Where a mean over all sensors for each type was calculated and the corresponding standard deviation was included. The standard deviation is an effective metric for showing the variation in performance of low-cost sensors of the same type.

When the LCS nodes were not used for the campaign, they were located on the rooftop of the neighboring building to the air quality monitoring station, HCØ. After calibration, the O_3 and NO_2 readings from the LCS were adjusted with field offsets determined from the difference between the LCS readings obtained during nighttime at the rooftop and the readings from the reference instruments at HCØ. The $PM_{2.5}$ values were adjusted with a field offset determined from the difference between the readings obtained during nighttime at the rooftop and the readings from the TEOM instrument at HCAB. The LCS data was screened for outliers, which were then removed. The outliers were defined as concentrations above 10 times the standard deviation of the concentration, which corresponded to less than 2 % of data for each LCS.

The LCS readings during the campaign were adjusted using a generalized additive model (GAM) (Wood and Scheipl, 2020) to more accurately determine the exposure levels. The time of day and the day of the week were included as factors in the GAM, together with temperature, relative humidity, wind speed, and wind direction.

Statistical analysis was performed using the two-sample t-test, comparing the means of two samples to find the differences among measured pollutant concentrations between the two routes. Also, a pairwise t-test was used to calculate multiple t-tests between all possible combinations of groups to compare the sections of the lake. The statistical significance was assumed at the 5 % level ($p < 0.05$). Extremely small p-values are rounded down to 0.

Within the study, we used data from an accelerometer (x, y, z) and a gyroscope (x, y, z). For the analysis, the orientation-independent magnitude ($M(t)$) of the accelerometer and gyroscope vectors are used:

$$M(t) = \sqrt{s_x(t)^2 + s_y(t)^2 + s_z(t)^2} \quad (3)$$

where s_x, s_y, and s_z are the three axes of the accelerometer and gyroscope, respectively.

3. Results and discussion

3.1. Calibration and intervariability

The results of the validation of LCS node calibration are shown in Table 2, whereas the results from the calibration training period are found in Table 3 in the Appendix. As shown in Table 2, the LCS measurements of O_3 performed better than the LCS measurements of NO_2 and $PM_{2.5}$ (TEOM) in terms of size of R^2, slope difference from 1, intercept difference from 0, and size of RMSE, MBE, and MAE. For O_3 and NO_2, the ANN model was the least successful, followed by the LR model, with the MLR model being the most effective when testing the calibration models in the validation period. All statistical indicators follow this pattern, except for the slopes of the ANN model for NO_2 and O_3 (0.85 and 0.99), which is acceptable and similar to the LR and MLR models. As expected, the low-cost sensors generally performed better in the training period, but in particular, the ANN models struggled to predict the observed concentrations in the validation period. Machine learning models such as ANN typically require larger datasets for training. Otherwise, they risk overfitting to the training periods, which may explain the poor performance of the ANN (Giordano et al., 2021).

Regarding the optical PM sensor, two $PM_{2.5}$ higher-grade instruments were available at HCAB: TEOM and LVS. The PM sensors were calibrated against both instruments to further detail sensor performance. Note that the TEOM instrument is not accepted as a designated reference or field equivalent method (EPA, 2022), but is a 'higher-grade' instrument that can be compared with low-cost sensors at a greater time resolution than LVS. The optical PM sensor inside Gen4P16 had a malfunction and was not included in the analysis. Generally, calibration based on LVS yields the highest R^2 and the lowest RMSE, MBE, and MAE. However, when using LVS for calibration, all calibration models overestimate the $PM_{2.5}$ concentration (slopes = 1.9, 1.6, and 1.2) and have low intercepts (<-3.8). On the other hand, calibration based on TEOM shows that only the LR model overestimates the concentration (slope = 2.0) of $PM_{2.5}$, while the ANN model underestimates it. Note that for $PM_{2.5}$ calibration with LVS as the reference, the R^2-values are higher during the validation period than in the training period, primarily due to the relatively smaller amount of validation data.

The MLR models outperform the LR models since they include the environmental parameters, *T* and *RH*. It is known that optical PM sensors tend to give high readings relative to TEOM and LVS at *RH* levels above 70 % as water condenses on the particles and/or due to changes in light intensity caused by the humidity in the sampling system (Chatzidiakou et al., 2019; Vogt et al., 2021). In addition, high relative humidity affects the accuracy of TEOM measurements, since the heat flow inside the TEOM dries the particles, causing an underestimation of PM mass concentrations (Wanjura et al., 2008; Allen et al., 1997).

During the co-location, the mean *RH* was 69.6 % (range = 33.3–95.0 %) and the mean *T* was 20.5 °C (range = 15.1–29.9 °C). While all campaign measurements were collected in the daytime, the calibration period also included nighttime measurements. However, the *RH* and *T* levels were only slightly lower during the measurement campaign relative to the co-location period with a mean *RH* of 61.5 % (range = 38.8–90.3 %) and a mean *T* of 18.0 °C (range = 8.9–29.6 °C).

For the campaign, we choose to proceed with MLR as the calibration model, as it was the most successful across all pollutants and is a simple and straightforward model. As for the calibration of $PM_{2.5}$, we proceeded with TEOM-based calibration coefficients since MLR calibration based on LVS yielded high slopes (1.6 ± 0.1) and low intercepts (−7.6 ± 1.8 μg m^{-3}).

The intervariability between the low-cost sensors and comparison with the references during the MLR calibration training and

Table 2
Statistics for the linear regression (LR), multivariate linear regression (MLR), and artificial neural network (ANN) models trained in the first three weeks and validated on the last week of the calibration period. LVS (Low volume sampler) and TEOM (tapered element oscillating microbalance) in parentheses illustrate the higher-grade instrument used for the PM sensor calibration. The performance metrics are based on the validation period and calculated from 1-h averaged NO_2 and O_3 data, and 3-h averaged $PM_{2.5}$ (TEOM) and 1-day averaged $PM_{2.5}$ (LVS) data. The statistical indicators, intercept, root-mean-square error (RMSE), mean bias error (MBE), and mean absolute error (MAE), are in ppb for NO_2 and O_3, whereas they are in μg m^{-3} for $PM_{2.5}$. All values are means over all sensors with their corresponding standard deviation of the results shown in parentheses.

Pollutant	Model	R^2	Slope	Intercept	RMSE	MBE	MAE
				ppb/μg m^{-3}	ppb/μg m^{-3}	ppb/μg m^{-3}	ppb/μg m^{-3}
NO_2	LR	0.62 (0.12)	0.82 (0.19)	2.4 (4.5)	7.1 (1.0)	1.2 (1.5)	5.6 (0.68)
	MLR	0.64 (0.05)	0.86 (0.09)	2.4 (3.0)	7.0 (0.40)	0.22 (1.6)	5.4 (0.47)
	ANN	0.44 (0.16)	0.85 (0.31)	−9.0 (10.0)	8.6 (1.3)	14.0 (7.1)	15 (6.0)
O_3	LR	0.77 (0.08)	0.96 (0.09)	0.2 (3.0)	3.3 (0.51)	0.90 (1.1)	2.8 (0.44)
	MLR	0.79 (0.06)	0.94 (0.1)	1.2 (3.0)	3.1 (0.41)	0.42 (1.3)	2.7 (0.44)
	ANN	0.73 (0.16)	0.99 (0.15)	−3.8 (3.8)	3.7 (0.98)	4.2 (2.4)	5.2 (1.5)
$PM_{2.5}$ (TEOM)	LR	0.41 (0.05)	2.0 (0.26)	−9.5 (2.0)	2.9 (0.12)	0.27 (0.27)	2.36 (0.08)
	MLR	0.48 (0.01)	1.0 (0.03)	−2.0 (0.5)	2.7 (0.04)	1.2 (0.31)	2.3 (0.09)
	ANN	0.50 (0.02)	0.7 (0.03)	0.89 (0.27)	2.6 (0.04)	2.6 (0.29)	3.0 (0.22)
$PM_{2.5}$ (LVS)	LR	0.90 (0.04)	1.9 (0.18)	−8.7 (1.3)	0.87 (0.17)	0.15 (0.41)	1.3 (0.09)
	MLR	0.96 (0.05)	1.6 (0.1)	−7.6 (1.8)	0.51 (0.28)	1.3 (0.52)	1.63 (0.34)
	ANN	0.73 (0.07)	1.2 (0.12)	−3.8 (1.3)	1.5 (0.17)	1.9 (0.16)	2.1 (0.15)

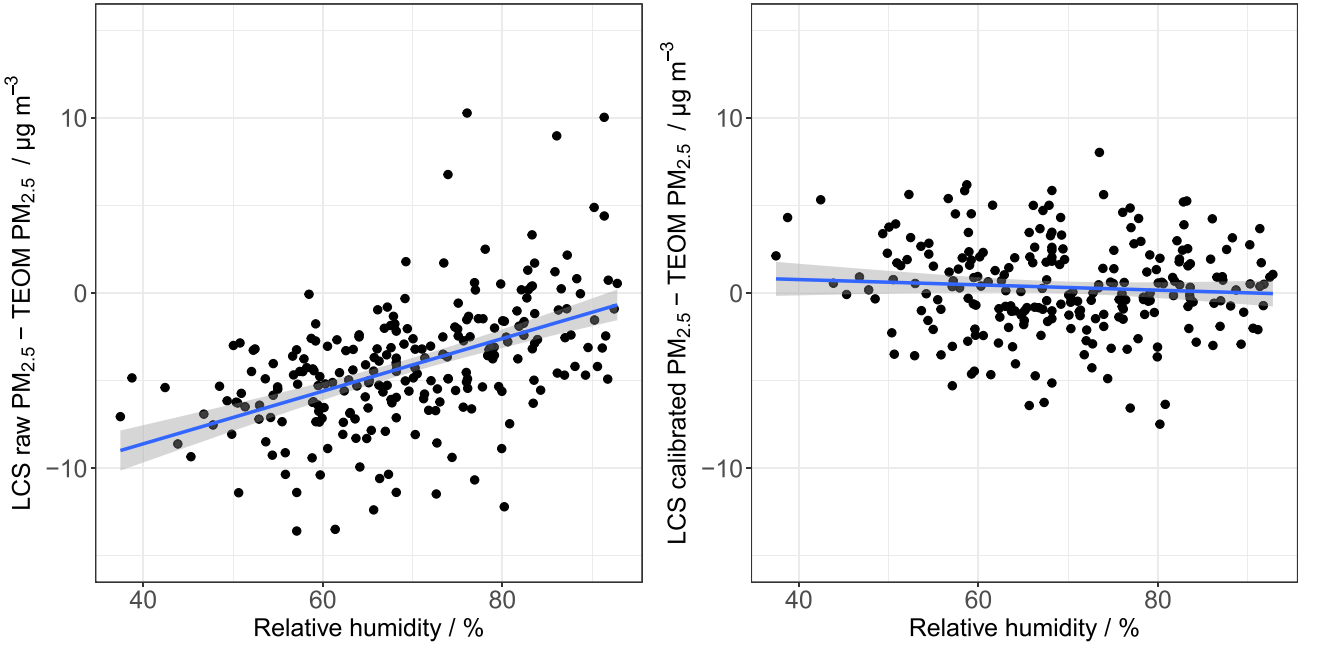


Fig. 2. Example of one representative SDS011 sensor in a Gen4P node. Difference between the LCS raw $PM_{2.5}$ concentration and the TEOM $PM_{2.5}$ concentration as a function of relative humidity (left), and the difference between the LCS MLR calibrated $PM_{2.5}$ concentration and the TEOM $PM_{2.5}$ concentration as a function of relative humidity (right). A linear regression line is added (blue line with shaded 95 % confidence interval) in both panels. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

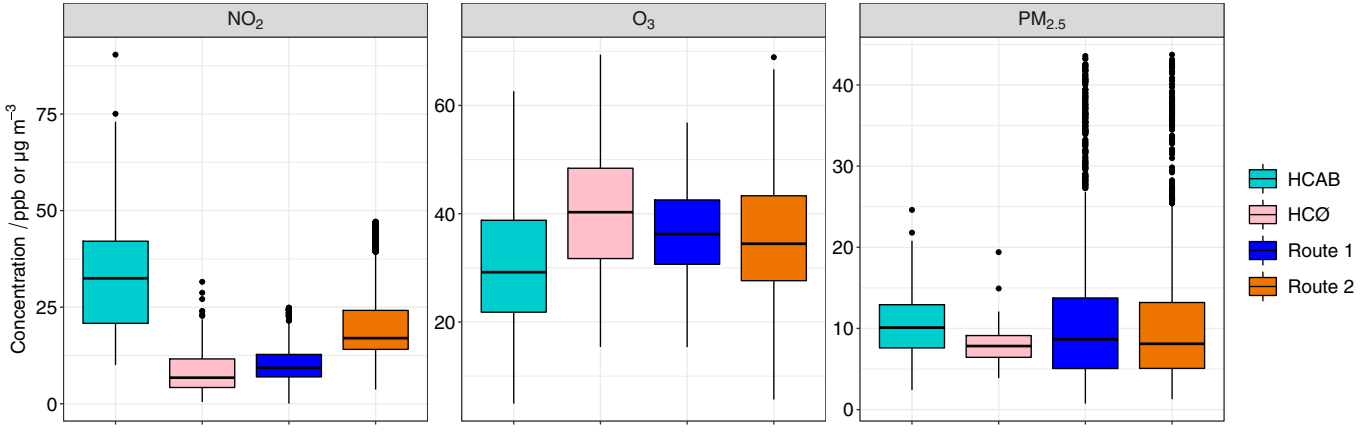


Fig. 3. Box plot of NO_2 (left), O_3 (middle) and $PM_{2.5}$ (right) for Route 1 (blue), Route 2 (orange), HCØ (pink) and HCAB (turquoise) over the entire measurement period. The concentration of NO_2 and O_3 are in ppb, whereas the concentration of $PM_{2.5}$ is in µg m^{-3}. Each box is a mean of all Gen4P nodes on the respective backpack. Note the different axis scales. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

validation periods are shown in Figs. A6–A9 in the Appendix. The lowest intervariability was seen between the optical PM sensors. Comparing the calibrated $PM_{2.5}$ sensor measurements over the validation period, a mean R^2 of 0.98 ± 0.014 was recorded with a time resolution of 3 h, whereas a mean R^2 of 0.94 ± 0.063 was seen with a time resolution of 24 h. The mean R^2-values are calculated across all sensors, with the corresponding standard deviation provided after the mean. Compared to the PM sensors, there is a higher intervariability for the NO_2 and O_3 sensors. However, the intervariability is still relatively low, with mean R^2 of 0.78 ± 0.071 for NO_2, and 0.83 ± 0.10 for O_3, during the validation period with a time resolution of 1 h. Note that the intervariability might be slightly lower for the PM sensors in part due to the lower time resolution.

As mentioned previously, low-cost PM sensors can have a strong *RH* dependence. The difference between $PM_{2.5}$ concentrations obtained by the low-cost PM sensor and TEOM (before and after calibration for one PM sensor) as a function of relative humidity is shown in Fig. 2. This plot is representative of all the PM sensors used in this study. The difference between the low-cost PM sensor and

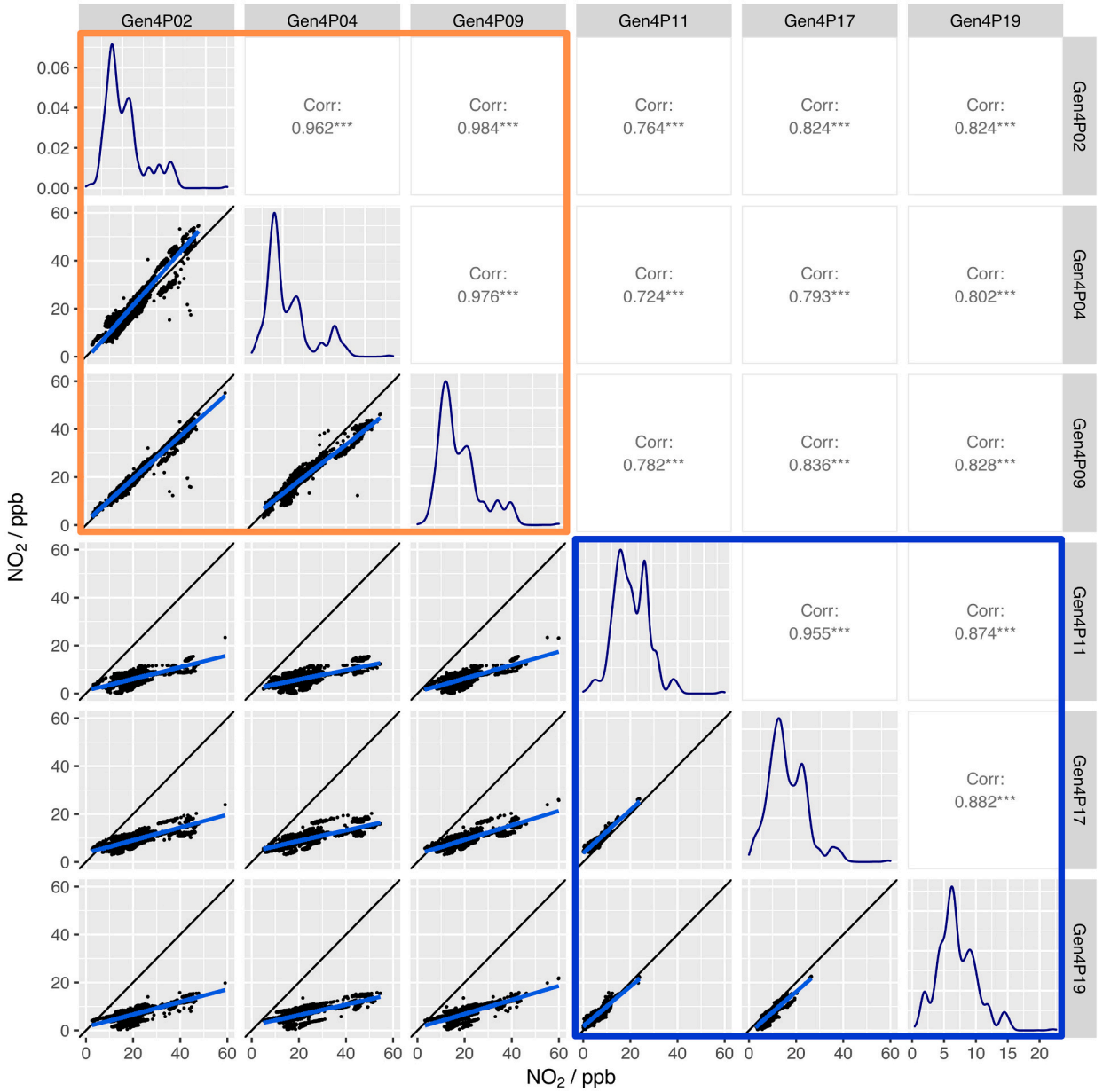


Fig. 4. Correlation matrix of all NO_2 sensors' adjusted calibrated readings with a 1-min time resolution during the campaign, displaying scatter plots (bottom left), density diagrams (diagonal), and pairwise Pearson's correlation coefficients (top right). All values are in ppb, except the *y*-axis of the density diagrams, which is the density of probability. Note the different scales. The orange and blue boxes illustrate on which backpacks the LCS nodes were located, where the orange box illustrates Route 2 and the blue box illustrates Route 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

TEOM is slightly dependent on the relative humidity, with an $R^2 = 0.25$. However, after the MLR calibration, with *RH* included as an input variable, the *RH* dependence disappears, resulting in a decreased difference between the calibrated sensor readings and TEOM.

3.2. Exposure levels

Fig. 3 shows the concentrations of NO_2, O_3, and $PM_{2.5}$ on Route 1 and Route 2 for the entire campaign, along with the concentrations measured by the higher-grade instruments at HCAB and HCØ during the sampling period. Summary statistics for GAM-adjusted data obtained in the campaign are shown in Table A4 in the Appendix. As expected, the NO_2 concentrations on both routes are higher than the urban background concentration measured at HCØ. However, the concentrations on the routes are lower than the concentration found at HCAB. Greater concentrations of NO_2 are seen on the route closest to the road (Route 2) and this is significantly different than the mean of Route 1 ($p = 0$). The NO_2 concentrations found at Route 1 and 2 were significantly different from the urban background at HCØ with *p*-values of 0 and 0.009 for Route 2 and 1, respectively. Likewise, the concentrations were significantly different from the urban roadside station at HCAB (both *p*-values = 0). The concentration of O_3 was higher at the two routes relative to the concentration found at HCAB, but lower than the one at HCØ. The O_3 concentration was significantly different for the two routes compared to HCØ (both *p*-values <0.0004) and HCAB (both *p*-values <0.0002), but the two routes were not significantly different from one another ($p = 0.43$). The concentration of $PM_{2.5}$ was also similar for the two routes, with no significant difference between the routes ($p = 0.98$), HCAB ($p = 0.071$), or HCØ ($p = 0.065$).

These relative concentrations match what we would expect, based on the sources and atmospheric chemistry involved, as NO_2 is a local pollutant, produced through traffic emissions with O_3 being consumed in the process, whereas the $PM_{2.5}$ concentration has more of a regional character.

The effect of covariates on the exposure levels is investigated by examining the summary of the GAM models. This reveals the influence of different parameters on pollutant levels. The cumulative deviance explained by the different parameters is shown in Table A5 in the Appendix. The simplest models, adjusted only for the time of day and day of the week, explained 5.4 ± 2.0 % for NO_2, 9.2 ± 2.6 % for O_3, and 11.4 ± 1.4 % for $PM_{2.5}$, where the percentages are the mean over all Gen4P nodes, with their corresponding standard deviation. While the adjustment for wind speed and direction combined explained about 38 % of variance for NO_2, 14 % for O_3, and 19 % for $PM_{2.5}$, the adjustment for temperature and relative humidity combined explained about 10 %, 55 %, and 15 % for NO_2, O_3, and $PM_{2.5}$, respectively. When analyzed individually, wind direction had the strongest influence on the NO_2 levels (33 %), followed by wind speed (17 %). For O_3, the temperature had the strongest influence (35 %), and for $PM_{2.5}$, temperature and wind direction were the strongest influencers at 13 and 12 %, respectively.

These results show the validity of the analysis as the explained deviance follows the expected patterns. Concerning NO_2, the most influential predictors are wind direction and wind speed, highlighting the significance of dilution and transport processes (Carslaw et al., 2007). Meanwhile, the primary predictor of O_3 is temperature; aligning with the photochemical production of O_3 and the fact that the reaction between NO_2 and O_2 to form O_3 is particularly evident in the presence of sunlight. This also explains the observed temperature effects on the NO_2 concentrations. These findings align with previous studies by Pearce et al. (2011) and Elminir (2007). For $PM_{2.5}$, the strongest predictors were temperature and wind direction, aligning well with the findings of a study by Tai et al. (2010). They observed a positive relationship between temperature and $PM_{2.5}$ and suggest that this correlation could be explained by temperature's association with stagnation and fronts. Compared to $PM_{2.5}$ and O_3, NO_2 exhibits less variance explained by meteorology, which reflects the impact of local emissions and short-range transport.

3.3. Comparison between sensors

Each backpack contained two types of PM sensors (SPS30 and SDS011) in addition to MOx NO_2 and O_3 sensors (MiCS-6814). All readings are corrected with GAM as described in the data analysis section. Fig. 4 shows the intervariability for the NO_2 sensors used in the campaign. The orange and blue boxes illustrate which backpacks the LCS nodes were located on, with the orange box representing Route 2 and the blue box representing Route 1. There is a low variability between the MOx NO_2 sensors located on the same backpack ($R^2 = 0.88 \pm 0.09$, and slope = 0.97 ± 0.00), whereas the variability is higher between the MOx NO_2 sensors located on different backpacks ($R^2 = 0.64 \pm 0.06$, and slope = 0.29 ± 0.02). This is associated with the spatially variable nature of NO_2. In contrast, there is almost no variation between the O_3 levels obtained from the MOx sensor even if they are located on different backpacks ($R^2 = 0.85 \pm 0.04$ for different backpacks compared to $R^2 = 0.90 \pm 0.02$ on the same backpacks), as seen in Fig. A10 in the Appendix.

A correlation matrix for all PM sensors used in the study is shown in Fig. A11 in the Appendix. There is generally low variation between all PM sensors, independent of backpack location, with mean R^2 of 0.94 ± 0.02 for SDS011 and mean R^2 of 0.84 ± 0.01 for SPS30. However, when comparing the two different types of low-cost PM sensors, a mean R^2 of 0.78 ± 0.03 was obtained. In the PM sensor correlation matrix the sensors are grouped based on sensor type, rather than backpack location, as this was seen to have a larger difference. A previous study by Bulot et al. (2020) found that SDS011 sensors can have a significant and variable time delay, which can range from 1 to 100 s and vary between units and experimental conditions. This time delay may explain the measurement points outside of the trend. Decreasing the time resolution can potentially overcome this delay and result in an improved correlation between the sensors, as seen in Figs. A8 and A9 in the Appendix. The latter also applies to the MOx NO_2 and O_3 sensors, as seen in Figs. A6 and A7 in the Appendix.

We also investigated whether the accuracy of the LCS data was dependent on speed. To do this, we examined the difference between the sensors located on the same backpack as a function of speed. During the campaign measurement, the mean speed was 4.5 km hour^{-1} (range = 0 to 6.5 km hour^{-1}). The difference between the sensor readings did not correlate with speed, indicating that the relative values were not

affected by the movement, but the absolute values could be systematically affected for all sensors at different speeds.

3.4. Additional sensors for estimation of mode of deployment

It is imperative to know the location of the participants in personal exposure monitoring studies. GPS data, combined with additional sensors such as temperature, humidity, accelerometer, gyroscope, and light sensors, can be used to classify micro-

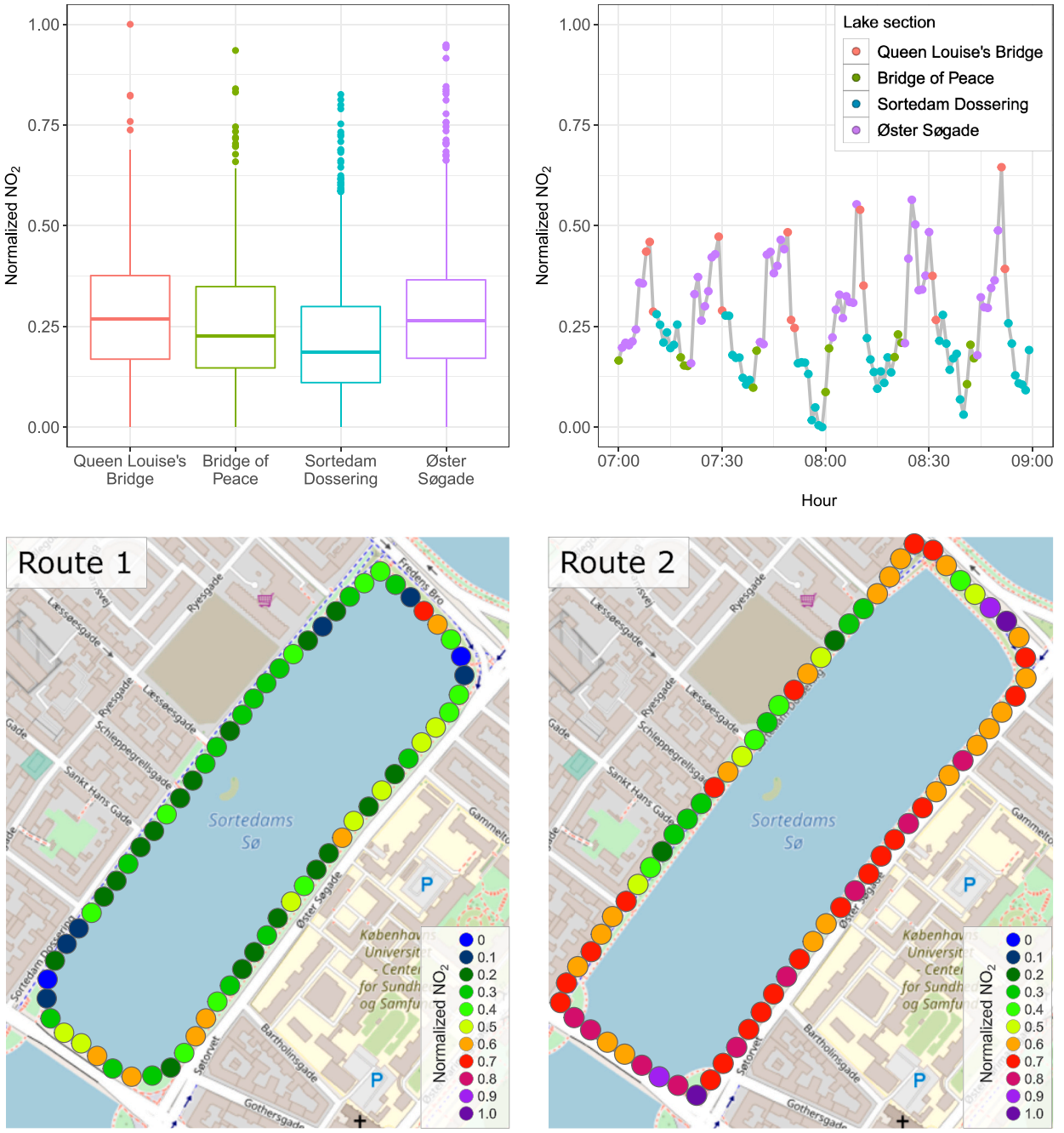


Fig. 5. Upper Panel: Box plot showing normalized calibrated NO_2 values for the complete campaign (left) and example time series for June 18, 2021, in the morning (right). Lower Panel: Normalized spatially averaged NO_2 for Route 1 (left) and Route 2 (right) on June 18, 2021. Note that the normalization in the upper panel is done for the complete campaign, whereas for the lower panel, it is normalized with respect to the example day for clarity. The maps are obtained from © OpenStreetMap contributors (OpenStreetMap contributors, 2021). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

environments and modes of deployment (Russell et al., 2022b). Each backpack contained multiple accelerometers, gyroscopes, and light sensors. An example from one representative PEM node's accelerometer, gyroscope, and light sensor during the campaign is seen in Fig. A12 in the Appendix. Only a single channel from the light sensor and the orientation-independent magnitude of the accelerometer and gyroscope readings are used. From this, the readings can be categorized based on the mode of deployment, distinguishing whether the LCS nodes were stationary or carried by volunteers while walking around the lake.

A simple machine learning model was developed to demonstrate the value of these additional low-cost sensors as an add-on for personal exposure monitoring studies. Random Forest was used to predict whether the LCS nodes were stationary or mobile, utilizing accelerometer, gyroscope, and light sensor data as input variables with 10-fold cross-validation. The results showed that 97.4 % of the instances were classified correctly, with a relative error of 14.7 % and a root relative squared error of 38.6 %.

3.5. Low-cost sensors for fine spatial resolution measurements

We investigate how the NO_2, O_3, and $PM_{2.5}$ concentrations vary with location around the Black Pond Lake. As previously mentioned, Black Pond Lake is a rectangular lake with four straight sections: Bridge of Peace, Queen Louise's Bridge, Øster Søgade, and Sortedam Dossering, located north, south, east, and west of the lake, respectively. The calibrated NO_2 values measured by Gen4P02 on Route 2 for each measurement period are shown in Fig. A13 in the Appendix. For the calibrated NO_2 data, there was higher variance between the different days than between the different sections of the lake. Therefore, to quantify the differences between the sections, the minimum value within each measurement period was subtracted from the respective measurement period, resulting in all data being in the same range. The same was done for O_3 and $PM_{2.5}$. Afterward, the NO_2, O_3, and $PM_{2.5}$ data were min-max normalized so each value is transformed into a decimal between 0 and 1 since the absolute values were not relevant for this analysis.

The normalized NO_2 values were categorized into the four different sections. A box plot for the whole campaign, split between the different sections, and an example of a measurement period are shown in Fig. 5. The lower panel of Fig. 5 shows spatially averaged normalized NO_2 concentration for Route 1 and Route 2 on one example date (June 18) to illustrate the relative differences between the two routes and the four sections of the lake. However, the example time series of the NO_2 concentration during a single measurement period, colored by section of the lake, clearly illustrates how the low-cost NO_2 sensors capture the difference between the different sections of the Black Pond Lake, as each lap is visible within the LCS data.

The mean normalized NO_2 concentration at Sortedam Dossering during the whole campaign was the lowest of the four sections, with a mean normalized NO_2 concentration of 0.21, compared to 0.28, 0.26, and 0.28 for Queen Louise's Bridge, Bridge of Peace, and Øster Søgade, respectively. A pairwise *t*-test showed that the NO_2 concentration at Sortedams Dossering was significantly different from the other three sections (all *p*-values below 0.05), while none of the other sections was significantly different from each other. A similar analysis was performed for O_3 and $PM_{2.5}$. However, for $PM_{2.5}$, there was no significant difference between the four sections of the lake. For O_3, the concentration at Sortedams Dossering was slightly higher than the other sections and significantly different from Øster Søgade ($p = 0$) and Queen Louise's Bridge ($p = 0.015$) but not Bridge of Peace ($p = 0.3$). This inverse relationship with NO_2 is expected since O_3 reacts with traffic-induced NO, forming NO_2.

For the NO_2, O_3 and $PM_{2.5}$ concentrations, the dependence on wind direction is shown in Figs. A14, A15 and A16 in the Appendix. For $PM_{2.5}$ no clear relationship with wind direction was seen. The mean NO_2 concentration at Sortedam Dossering is highest when the wind is coming from the southeast (112.5°–157.5°), which is when the wind is coming directly from Øster Søgade and crosses the lake. When the wind was coming from the north (337.5°–360° and 0°–22.5°), the highest NO_2 levels were seen at the Bridge of Peace, and when the wind was coming from the south (157.5°–202.5°), the highest concentrations were seen at Queen Louise's Bridge. This suggests that when the wind was perpendicular to the walking route, it could transport NO_2 toward the persons with the backpacks, resulting in the highest NO_2 concentrations. In comparison to NO_2, the concentration of O_3 did not exhibit as much variation dependent on wind direction. However, the concentration was generally highest at Sortedam Dossering, suggesting minimal traffic-induced NO that can consume O_3. During southerly winds, the O_3 concentration was lowest at Queen Louise's Bridge, showing the opposite pattern to that of NO_2 for this wind direction.

4. Conclusions

In conclusion, low-cost air pollution sensor (LCS) nodes are effective for assessing exposure to NO_2, O_3, and $PM_{2.5}$ in specific micro-environments. The study demonstrated that low-cost sensors can identify exposure levels in an urban environment with fine spatio-temporal resolution and highlighted the utility of LCS nodes for personal exposure monitoring. The study found a factor of two difference in NO_2 concentration levels between two routes, located on average 8 m apart in Copenhagen, Denmark. However, this variation was not found for O_3 and $PM_{2.5}$, which is expected since NO_2 has more local sources compared to O_3 and especially $PM_{2.5}$. This study shows that LCS nodes can be used to identify exposure differences or hotspots in areas that otherwise would not be measured. Overall, the study provides valuable insights into the potential of LCS nodes for air pollution monitoring. For future work, it would be interesting to compare modeled pollution levels along the same sampling path with the measured data.

Author statement

Louise Bøge Frederickson: Conceptualization, Methodology, Data Curation, Validation, Software, Formal analysis, Investigation, Writing - Original Draft, Visualization, Writing - Review and Editing.

Hugo Savill Russell: Conceptualization, Methodology, Data Curation, Validation, Software, Investigation, Visualization, Writing -

Review and Editing.

Dafni Fessa: Data Curation, Writing – Review and Editing.

Jibran Khan: Data Curation, Software, Writing - Review and Editing.

Johan Albrecht Schmidt: Resources, Software, Supervision, Writing - Review and Editing.

Matthew Stanley Johnson: Resources, Validation, Supervision, Funding acquisition, Writing - Review and Editing.

Ole Hertel: Conceptualization, Methodology, Resources, Validation, Supervision, Project administration, Funding acquisition, Writing - Review and Editing.

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All authors have read and agreed to the published version of the manuscript.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Louise Bøge Frederickson reports a relationship with DevLabs/AirScape that includes: funding grants. Hugo Savill Russell reports a relationship with DevLabs/AirScape that includes: funding grants. Johan Albrecht Schmidt reports a relationship with DevLabs that includes: board membership, employment, and equity or stocks. Matthew Stanley Johnson reports a relationship with DevLabs/AirScape that includes: board membership, employment, and equity or stocks.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.uclim.2023.101684.

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